

Intelligent Log Forensics

Abstract:

Modern web applications generate large amounts of logs from API requests, authentication services, frontend errors, backend failures, server operations, and performance activities. These logs contain important information about application issues, unauthorized access attempts, repeated API failures, slow endpoints, and possible security risks. However, raw logs are often unstructured and difficult to analyze manually. Intelligent Log Forensics is a lightweight system that helps analyze application logs and convert them into meaningful operational and security insights. The system accepts log files in different formats such as CSV, JSON, TXT, and LOG. It parses and normalizes the logs, checks data quality, extracts useful features, and classifies events into safe, warning, operational risk, or security risk categories. For security-related events, the system maps suspicious patterns to selected MITRE ATT&CK tactics and techniques. It also generates risk scores, incident timelines, explanations, suggested fixes, dashboard analytics, and downloadable forensic reports. This system helps developers, students, and small teams understand application failures and possible security issues without using complex enterprise monitoring tools.

Tech Stack :

- **Frontend:** HTML, CSS, JavaScript, Flask Jinja2
- **Backend:** Python (Flask)
- **Database:** PostgreSQL
- **Data Processing & ML:** Pandas, NumPy, Regex, Scikit-learn
- **Visualization:** Chart.js
- **Report Generation:** ReportLab / WeasyPrint
- **Security Mapping:** Local MITRE ATT&CK JSON Mapping

